

Lahore College for Women University

Department: **Computer Science**

Paper: **Applied Physics**

Total Time: **1hours, 20mintues**

Paper Type: **Subjective Type**

Total Marks: **20**

095

Attempt all the questions.

Q 2. (a) There are two types of materials explain their elastic and inelastic limits also express what type of stress and strain is produced in the bodies with their young modulus expression.

(b) If a body having mass m and velocity v then what will be the momentum and the force applied on that body?

(c) There are three different displacements which a car covered in 20mintues are as $d_1 = 6m$, $d_2 = 8m$ and $d_3 = 5m$ with the following angles $\theta_1 = 40^\circ$, $\theta_2 = 30^\circ$ and $\theta_3 = 0^\circ$ respectively. Now find the net displacement in terms of their components.

Q 3. (a) If a body moving under the influence of gravity, then explain the Newtons law which is applied on that body

(b) A rocket is moving with some speed and make an angle θ with the earth, explain the flight of the rocket along with the path which is followed by the rocket and its maximum and minimum range.

(c) Two individuals A and B sliding a motor of weight 225kg with a displacement d of magnitude 8.50m. the push F_1 of A is 12N at an angle of 30° downward, while the pull F_2 of B is 10N at 40° above the horizontal. What is the net work done on the motor by force F_1 and F_2 during the displacement d ?